SimuPharma

India's first AI-powered virtual pharmaceutical laboratory

₹2 Crore lab ← → ₹199/month

- 7 Real Physics-Engine Simulators
- Agentic AI Teacher
- Built with ₹0 Investment

SPPU STARTUP OLYMPIAD 2026 · WADHWANI FOUNDATION

Pranav Dhanwate · Founder & Developer

Would you board a plane if the pilot only trained on textbooks?

That's exactly how India trains its pharmacists.
2 lakh graduates every year — testing the medicines you take —
most have **never independently operated** the instruments they'll use on Day 1.

Colleges own ₹2 Crore in lab instruments.
Each student uses them 4 times in 4 years.

30:1 student-to-instrument ratio · 5 min per turn · batch sizes of 120+ · broken/outdated equipment

3,079	7.8 Lakh	90%	₹2 Cr
PHARMACY COLLEGES	STUDENTS ENROLLED	NEVER OPERATED SOLO	EQUIPMENT PER COLLEGE

THE SOLUTION

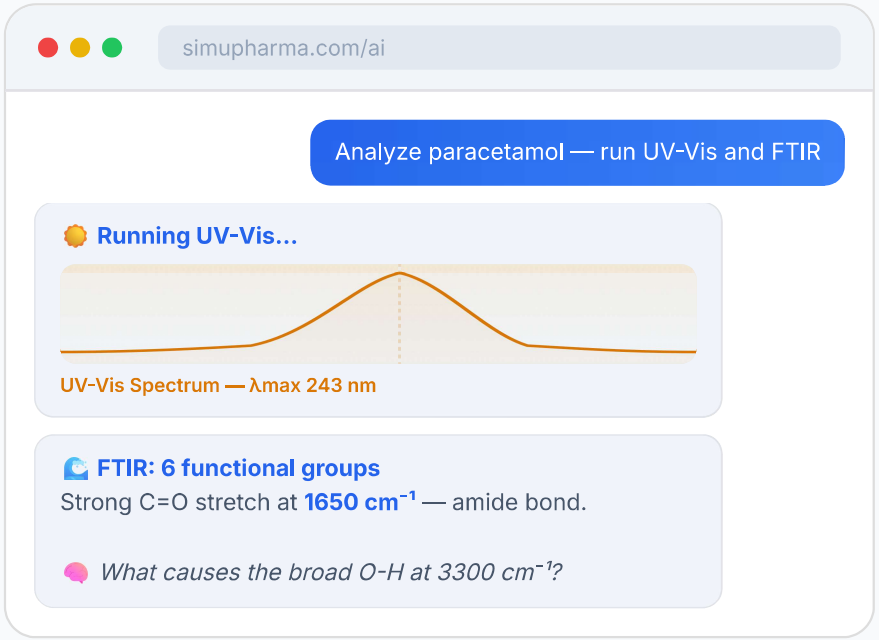
A ₹2 Crore lab in every student's pocket.

For ₹199/month.

7 real physics-engine simulators — not animations. Change flow rate → retention times shift. Change concentration → absorbance follows Beer-Lambert law.

+ An **AI teacher** that speaks, runs experiments, shows live spectra inline, and teaches like a professor available 24/7.

7	24/7	5×
INSTRUMENTS	AI TEACHER	CHEAPER THAN LABSTER



MARKET OPPORTUNITY

₹18,000 Cr market.
Zero Indian players.

Labster (Denmark) raised **$100M** at **$30/student/month**. India has 3,079 pharmacy colleges — and no local alternative. We're **Labster for India** at 1/10th the price.

- Global virtual lab: **$6.3B by 2030** — Grand View Research
- India pharmacy: **3,079 colleges, 7.8L students** — PCI
- India EdTech: **15% CAGR** — RedSeer
- NEP 2020 mandates **digital-first lab access**

TAM · TOTAL ADDRESSABLE MARKET

₹18,000 Cr

Global virtual lab + EdTech simulation market

SAM · SERVICEABLE AVAILABLE

₹1,200 Cr

India pharmaceutical + chemistry education

IN

SOM · YEAR 3 TARGET

₹60 Cr

200 colleges + 10K Pro students

BUSINESS MODEL

Freemium SaaS.
92% gross margin.

Pure software — **zero reagents, zero maintenance**. Same model as Canva, Notion, Figma.

Student Free	3 instruments · 5 sims/day · Acquisition funnel	₹0
Student Pro	All 7 instruments · AI teacher · Certs	₹199/mo
Institution	Faculty dashboard · NAAC reports · LMS	₹50K–3L/yr

UNIT ECONOMICS

92% GROSS MARGIN	28.7× LTV / CAC
₹150 CAC (ORGANIC)	₹4,300 AVG LTV (PRO)

YEAR 3 TARGET

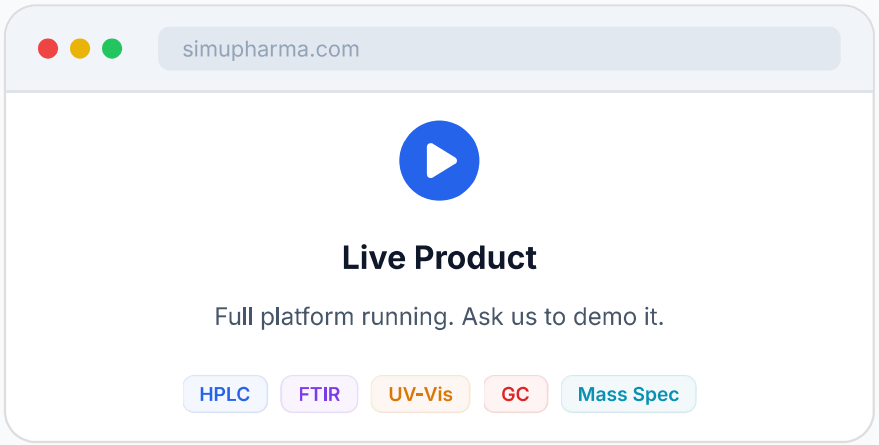
₹6 Cr
ARR Target

200 institutions × ₹1.8L avg
+ 10K Pro students × ₹199/mo

VALIDATION

Not a concept deck.
Built, live, demoable.

- **7 instruments live** — HPLC, FTIR, UV-Vis, GC, Mass Spec, Polarimeter, PSA
- **AI agentic teacher** — speaks, inline spectra, follow-up questions
- **Physics verified** — Beer-Lambert, Gaussian chromatography models
- **PCI curriculum mapped** — NEP 2020 aligned
- **Faculty dashboard** — analytics, NAAC/NBA reports
- **Payments live** — Razorpay subscriptions integrated
- **Gamification** — XP, streaks, badges, leaderboard, quizzes
- **3 college pilots** — discussions active, LOIs in progress



Live Demo Available

Ask us to run it right now. No slides — real product.

COMPETITIVE LANDSCAPE

Global giants, zero India presence.

FEATURE	SIMUPHARMA	LABSTER (DENMARK)	PRAXILABS (EGYPT)	PHYSICAL LAB
Price	**₹199/mo**	**$30/mo (~₹2,500)**	**$15/mo (~₹1,250)**	**₹2 Cr+ setup**
India Curriculum (PCI/NEP)	✓ **Mapped**	✗ Western only	✗ Generic	✓ **Manual**
AI Teacher (Agentic)	✓ **Speaks + runs expts**	✗ **None**	✗ **None**	✗ **Faculty-dependent**
Physics Engines	✓ **Beer-Lambert, Gaussian**	✓ **Comparable**	✗ Basic	✓ **Real instruments**
Inline Spectra/Charts	✓ **Real-time in chat**	✗ **None**	✗ **None**	✓ **On instrument**
Instruments	**7 pharma-specific**	200+ (generic science)	50+ (generic)	Varies by budget
NAAC/NBA Reports	✓ **Auto-generated**	✗ **No**	✗ **No**	✗ **Manual**
Language	**English + Hindi**	English only	English/Arabic	—

Our moat: PCI curriculum mapping + AI teacher + India-first pricing. Labster can't serve India at $30/mo. We own the ₹199 price point.

GO-TO-MARKET STRATEGY

Bottom-up adoption. Students first, institutions follow.

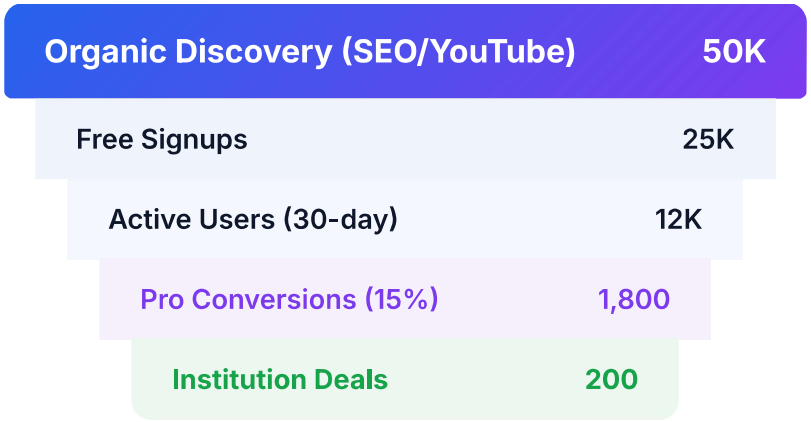
1. **Free tier virality**
   Students discover SimuPharma via Google/YouTube (pharmacy students actively search "HPLC simulator"). Free tier → organic viral loop within college batches via WhatsApp groups.

2. **Pro conversion**
   ₹199/month — less than chai money. Convert power users through AI teacher value + quiz prep + certificates. Target: 15% free→Pro conversion.

3. **Institutional sales**
   When 30+ students in a college use SimuPharma, pitch the principal: "Your students already use us. Get the faculty dashboard + NAAC reports." Direct outreach + PCI conference presence.

4. **Government channels**
   AICTE/NEP digital lab mandates → state-level partnerships. Target: 5 state pharmacy council MOUs by Year 2.

ACQUISITION FUNNEL



₹150	30 days	90 days
CAC (ORGANIC)	SALES CYCLE (STUDENT)	SALES CYCLE (INST.)

THE TEAM

Built solo from scratch.
Backed by a strong team.

I didn't read about the 30:1 student-to-instrument ratio in a report — I lived it. Self-taught coder who built the **entire platform alone**, from frontend to AI, without spending a single rupee.

Pranav Dhanwate
PD
Founder · 3rd Year B.Pharmacy · Full-Stack Developer + AI
Self-taught coder through online courses. Built entire SimuPharma platform solo: 7 simulators, FastAPI backend, Gemini AI integration, Razorpay payments, gamification. Zero external funding used.

CORE TEAM

- AP Aditya Pol
- MK Mayur Kangane
- SS Satyajit Shelar
- TS Tejas Shirsath
- PS Piyush Salve

WHY US?

Founder = Target User
Pharmacy student who experienced the exact problems daily. Deep understanding of PCI curriculum, lab dynamics, and student needs.

Zero-Cost Execution
Full platform built with ₹0 investment — pure sweat equity. Self-taught coding, no outsourcing, no external help. Proves extreme resourcefulness.

Speed of Execution
7 instrument simulators, AI agent, full payment stack, gamification — all built as a student while attending classes.

FACULTY MENTOR

Dr. Suhas Siddheshwar
HOD, Pharmaceutics
Pravara Rural College of Pharmacy, Loni

FINANCIAL PROJECTIONS (3-YEAR)

Profitable by Year 2. ₹6 Cr ARR by Year 3.

YEAR 0 (NOW)	YEAR 1	YEAR 2	YEAR 3
₹0 spent	**₹20L**	**₹1.5 Cr**	**₹6 Cr**
Fully bootstrapped Pre-revenue **Zero investment**	500 Pro users 10 Institution pilots ₹150 CAC organic	3,000 Pro users 50 Institutions **Break-even**	10K Pro users 200 Institutions **Profitable**

KEY ASSUMPTIONS

- 15% free→Pro conversion rate (industry avg: 10-20%)
- ₹150 CAC through organic SEO + WhatsApp virality
- ₹1.8L avg. institution deal (weighted across Starter/Standard/Premium)
- 8% monthly churn for Pro (industry: 5-12%)
- 92% gross margin (pure SaaS, Gemini API cost ~₹2/user/mo)

PATH TO ₹100 CR

If we capture just **5% of India's pharmacy colleges** (154 colleges × ₹3L/yr) + **50K Pro subscriptions** → that's **₹16.6 Cr ARR**. Add chemistry departments (4,000+ colleges) and we're looking at a ₹100 Cr revenue business within 5 years.

THE ASK

Raising ₹15 Lakhs

Pre-seed · Built this far with ₹0

Built the **entire product with zero investment**. ₹15L accelerates us to **₹20L ARR in 12 months** and positions for a Seed round at ₹5 Cr+ valuation.

USE OF FUNDS



₹6L — Product
Mobile apps, 5 more instruments, cloud

₹4.5L — Marketing
SEO, YouTube, college activations

₹3L — Team
1 pharma content specialist

₹1.5L — Operations
Cloud infra, API costs, compliance

Built with ₹0 → Asking ₹15L
Imagine what we build with actual funding.

12-MONTH MILESTONES

Q1 — **Launch mobile apps**
iOS + Android · Mobile-first access

Q2 — **10 college pilots live**
Pravara + SPPU network

Q3 — **500 paying users**
Pro + Institution combined

Q4 — **₹20L ARR → Seed Ready**
Target ₹3-5 Cr raise

ROADMAP

What this funding de-risks.

M3 — M6 — M12

Month 3
Product Complete

Mobile apps launched
12 instruments total
Advanced AI teacher v2

Month 6
15 LOIs Received

College partnerships signed
500+ active users
First paying institutions

Month 12
₹20L ARR

Revenue milestone hit
Seed round ready
Chemistry vertical started

🎯 **Every rupee has a milestone attached.**

₹15L in → ₹20L ARR out in 12 months. That's a 1.3× return on pure operational spend — before any valuation growth.

VISION / CLOSING

Become India's leading virtual lab platform, powering 10 lakh students by 2030.

From pharmacy to chemistry, biology, engineering and medicine — every Indian student deserves a lab that's always open.

Let's talk:

simupharma.com

simupharm.team@gmail.com

9309398358

Want to see it live?
We can demo the full platform right now.